

Joshua Waters

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Profile

Software engineer with commercial experience across Python, Angular and TypeScript, and a systems engineering background from a BEng in Mechatronics, Robotics and Automation Engineering. Experienced in building production tools end-to-end, from PyQt6 GUI applications and signal processing pipelines to Angular dashboards and the backend services behind them. Seeking a software engineering role with a route into infrastructure, DevOps, and security, supported by a dissertation in cyber-physical systems security and hands-on Linux and self-hosting experience.

Key Skills

- **Programming Languages:** Python, JavaScript, TypeScript, PHP, HTML, SQL, CSS/SCSS.
- **Frameworks & Libraries:** Angular, RxJS, Angular Material, Pandas, NumPy, SciPy, SQLAlchemy, Matplotlib, Requests, PyQt6, Pytest, Jupyter Notebooks.
- **Databases:** MySQL, MongoDB.
- **Practices:** OOP, Agile development, REST API integration, Testing Frameworks, Version Control (Git, GitLab, GitHub), Signal Processing, Data Analysis.

Career Summary / Work experience

Aeromark, Bedford (2024 – Present)

Junior Software Engineer (2026 – Present)

- Lead developer on a production Angular/TypeScript IoT asset management dashboard, taking it from prototype to active internal deployment and growing it to 26 widget types, now owning the framework and the roadmap for configurable customer-facing layouts.
- Maintain and extend a portfolio of production IoT tooling day to day, across Angular, TypeScript, PHP, Python and MySQL, from feature work through to production support.

Technical Specialist (Software Engineer) (2025 – 2026)

- Joined the IoT team, building sensor tooling, data analysis pipelines, and frontend prototypes from the ground up.
- Took full ownership of half a dozen production Python tools following a colleague's departure, including hub status monitoring, hub programming, and other IoT utilities, maintaining minimal disruption across customer operations over approximately 12 months.
- Mentored a new Angular developer over 6 months across Angular frontend, PHP backend, SQL, and existing system architecture, accelerating onboarding and contributing efforts to decouple and modernise the codebase.

Key projects include:

- Sole developer of a sensor soak and calibration app built from the ground up over 3 months, incorporating multi-threading, PyQt6 GUI, connection pooling, SQL injection prevention, and an automated testing framework, replacing a predecessor with a high failure rate with a seamless, production-ready soak-to-calibration pipeline.
- Independently designed and built a signal-processing data analysis toolkit for IoT temperature monitoring, implementing fast Fourier transforms (FFT), moving averages, and anomaly detection within a dataset agnostic architecture, enabling defrost cycle analysis and predictive maintenance logic required to forecast asset servicing requirements.
- Built an asset health scoring pipeline on that toolkit, scoring refrigeration temperature series over rolling windows into a composite score weighing setpoint adherence, duration-aware breach compliance, and compressor cycle regularity derived from peak/trough detection and period variance; runs in parallel across the asset estate and persists per-window results for reporting.
- Designed and built the dashboard framework. Layouts defined as JSON and served from an API, decoupled widget and datasource registries so a widget is independent of where its data comes from, signals-based filters and drill-through detail views resolved at runtime.

Trainee Tech Support (2024 – 2025)

- Quickly became the go-to contact between development and PDM teams for technical queries across the full software stack.
- Took ownership of technical customer queries, using SQL to investigate and resolve backend database issues.
- Began developing Python tooling for temperature monitoring data analysis, forming the foundation of a signal processing toolkit I continued to develop in the Technical Specialist Role.
- Trained colleagues while still in own training period, recognised internally for fast knowledge acquisition.

Bar Staff, Bar One / Foundry at Students' Union, Sheffield (2021 – 2024)

- Balanced 18+ hours of work per week alongside a full-time degree, maintaining strong academic performance throughout.
- Took on managerial responsibilities beyond the role, training new staff and running bars independently.

Customer Sales Assistant, Stagedoor Harpenden (2016 – 2019)

- Solely managed the business for an extended period, trusted by the owner to handle day-to-day operations and financial management.

Voluntary Work

- Raised over £600 for Keech Hospice through organised fundraising efforts. (2015)
- Raised funds for Rethink Mental Illness through the Miles for Mental Health initiative. (2021)

Education

BEng Mechatronics, Robotics and Automation Engineering, University of Sheffield (graduated 2022) 2:1 with Honours Dissertation: 'Sparse Data Injection in Cyber-Physical Systems', an individual research project investigating security vulnerabilities in critical infrastructure systems.

Sir John Lawes, Harpenden - A Levels Maths (A), Physics (B), Computing (B); 11 GCSEs A* - C

Hobbies & Interests

Bouldering and rock climbing, running, gym, squash, guitar and music production, photography, model making, and spending time in nature.